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A New Six-Dimensional Hyper-Chaotic System

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***Abstract* – In the last 20 years, the chaotic system has been widely studied. A new six-dimension hyper chaotic system is introduced in this paper. We used a new chaotic system based on a six-dimension for the purpose of increasing chaos in the system, where the new system has twelve positive parameters, complicated chaotic dynamics behaviors and gives an analysis of the new systems. The basic characteristics and dynamic behavior of this system are investigated with a presence of chaotic attractor, dissipativity, symmetry, equilibrium points, Lyapunov Exponents, Kaplan-Yorke dimension , waveform analysis and sensitivity toward initial conditions . The results of the analysis exhibit that the new system contains three unstable equilibrium points and the six Lyapunov exponents. Maxim non-negative Lyapunov Exponent (MLE) is obtained as 1.04609 and Kaplan-Yorke are obtained as 4.00695, and the new system characteristics with , unstable , high complexity ,and unpredictability, The new system dynamics is simulated utilizing MATHEMATICA program . the phase portraits and the qualitative properties of the new hyper chaotic system have been described at the detail.**

***Index Terms*— chaotic, hyper, six-dimensional, unstable, waveform analysis, sensitivity to initial conditions.**

I. INTRODUCTION

The chaos in engineering systems has been studied along the last two decades [1] A chaotic system is a non-linear particular dynamic system, which has a number of features like sensitivity to initial conditions, no uniform and unpredictable behavior [2]. Moreover, the chaotic attraction has been seen as a chaotic group that tends to evolve dynamically [3] A chaotic system has a wide extent of applications in numerous Engineering and non-Engineering fields, a few of their applications are found in cleverly controls [4], the mathematics [5] , and the secure communication , etc. [6] Chaotic behavior obtains when the dynamical system develops with some particular values for parameters and initial conditions, the first system exhibits chaotic behavior discovered by accident in 1963 when professor Lorenz worked to build a model for weather paradigms, the mathematical model of the system is produced by three first-order differential equations with seven terms [7]. Various chaotic systems with three dimensions introduced such as the Rossler system,LÜ system , Jia system, and others. [8]. Because of the characteristics of the Chaos, it is implemented at present in many different fields such as engineering and computer science, physics, geology, mathematics robotics and etc. [9]. A new 6-D hyper-chaotic system with six state variables, twelve parameters, and six cross-product nonlinearities terms are introduced in this paper. The properties of the dynamic behavior of the new system are examined using the Mathematica programming language.

In this section, many researchers worked with the six-dimension chaotic system as below: Y. Zhao, et al. [10] designed a six-dimensional hyper chaotic System. the Lyapunov exponents are: LE1 = 0.5476, LE2 = 0.3223, LE3=-0.2143, LE4=-1.7447, LE5=-2.0578 and LE6=-88.5710 . Shibing Wang , Xingyuan Wang and Yufei Zhou [11] a novel complex Lorenz system with a flux-controlled memristor has been constructed , which can be expressed as a 6D (six-dimensional) real ordinary differential equations (ODEs), In order to reveal dynamical behaviors of the proposed memristor-based complex Lorenz system, we a_1=8 , a_2= 11, a_4 = 8 / 3 , $a = 0.67x\,10^{-3}$, β = $0.02x\,10^{-3}$, u(0) (2,0,1,4,0.1,0) and vary a_3 in range of [0, 200]. Pu Yue, Li Guodong and Zhao Jing [12] first Construct a generating chaotic sequence of six dimensional cellular neural network model, during calculating the Lyapunov index, the index values we can get : 6.2562, -0.9548, 0.6963, -13 .6470, -22.4462 . Lingzhi Yi et al. [13] a new six-dimensional hyperchaotic system is proposed and some basic dynamical properties including Some related characters are analyzed comprehensively, including symmetry, stability of equilibriums, dissipativity, bifurcations and Lyapunov exponent . When a = 10 , b = 8/3 , c = 28 , d = -1, e = 10 and r = 3 , the Lyapunov exponents are L1 = 0.362485 , L2 = 0.24709 , L3 = 0 , L4 = -0.225698 , L5 = -10.0017 , and L6 = -15.0708 .

II.CONSTRUCTION OF NEW SIX-DIMENSIONAL CHAOTIC SYSTEM

The hyper - chaos system has two or more of positive Lyapunov exponents in chaotic systems. Parameters and initial conditions are the basis upon which the chaotic sequences of the chaotic system, it is therefore difficult to predict their dynamic behavior and the attractive chaos is more complex. The hyper-chaotic has a distinct advantage System of excessive chaos on a low level of chaos. In this section, the new six-dimensional hyper chaotic system is the establishment then achieves to be hyper chaotic. The dynamic behaviors of a new system are obtained; this system can be obtained as following:

$$
\begin{aligned}
\dot{x} &= -ax + by + cw - dv \\
\dot{y} &= ex - fxz - ge^{v} \\
\dot{z} &= -hz + xy + iv \\
\dot{w} &= -w - yz - gv \\
\dot{v} &= x + jy - iz \\
\dot{u} &= kx - Lu - jzw
\end{aligned}
\qquad (1)
$$

Where x, y, z, w, v ,u and t $\in$ R+ called the states of system and a ,b , c , d , e , f , g , h ,i ,j ,k and l are the positive parameters of the system. The "(1) " shows the hyper chaotic attractors, when selected the system parameter values with the following determines the values: a$\in$[9,9.5] ,b$\in$[11,11.5] , c$\in$[0.1,0.5] , d$\in$[0.3,0.8] ,e$\in$[30,30.5], f=2.5 , g$\in$[5,5.5], h$\in$[3,3.5] , i$\in$ [0.5,1], j$\in$[4,4.5] , k$\in$[15,15.5] and l$\in$[25,25.5] and initial conditions as: x(0)=0.1 , y(0)=0.5, z(0)=3.5,w(0)=0.6. v(0)=0.4 and u(0)=0.1. The Figs. 1–8 illustrates the chaotic strange attractor in 3-D of the "(1)".

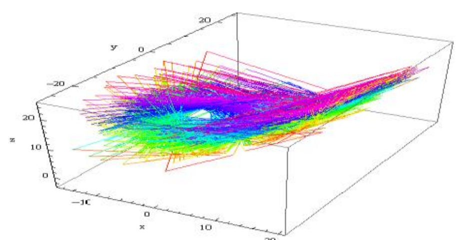

Fig.1. Chaotic attractors, three- dimensional view (x-y-z)

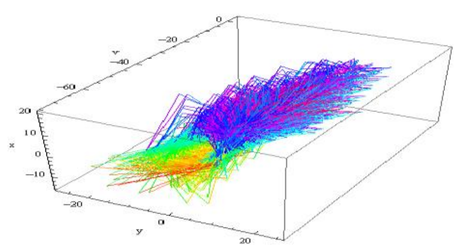

Fig.2. Chaotic attractors, three- dimensional view (x-y-v)

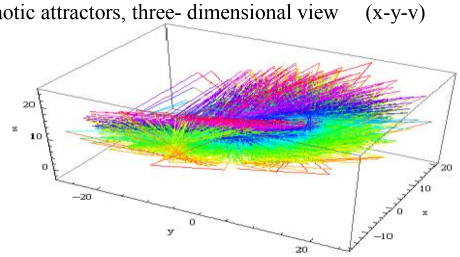

Fig.3. Chaotic attractors, three- dimensional view (z-y-x)

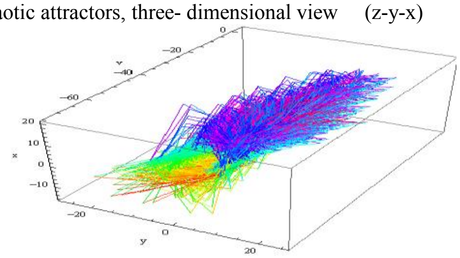

Fig.4. Chaotic attractors, three- dimensional view (v-x-y)

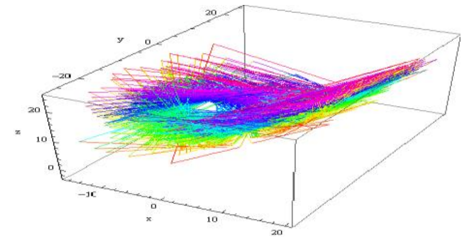

Fig.5. Chaotic attractors, three- dimensional view (y-z-x)

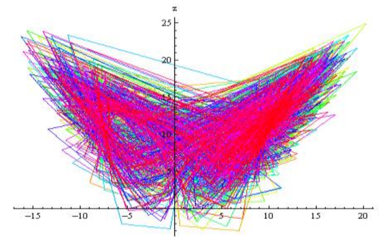

Fig.6. Chaotic attractors, *z* , *x* of novel system

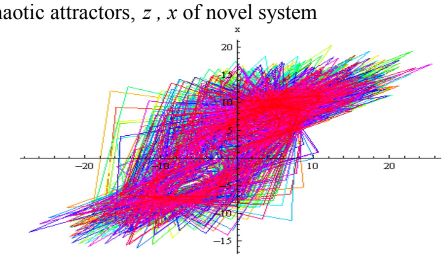

Fig.7. Chaotic attractors, *x* , y of novel system

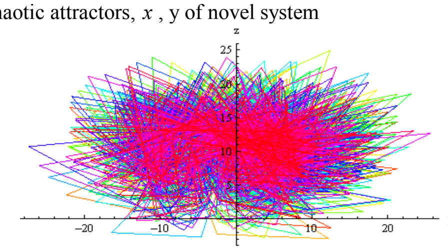

Fig.8. Chaotic attractors, z , y of novel system

III. DYNAMIC ANALYSIS OF THE PROPOSED CHAOTIC SYSTEM

The main properties and complex dynamics behavior of the new "(1)" is investigated in this section and it have the following basic Characteristics:

A. Dissipativity

" Equation (1)" can be described in vector form as:

$$
f = \begin{cases}
f_1 = \dfrac{dx}{dt} = -ax + by + cw - dv \\
f_2 = \dfrac{dy}{dt} = ex - fxz - g\,e^{v} \\
f_3 = \dfrac{dz}{dt} = -hz + xy + iv \\
f_4 = \dfrac{dw}{dt} = -w - yz - gv \\
f_5 = \dfrac{dv}{dt} = jy + x - iz \\
f_6 = \dfrac{du}{dt} = -ku - jzw + lx
\end{cases}
\qquad (2)
$$

The divergence of the vector field *f* on R^6 is given by:

$$\nabla \cdot f = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} + \frac{\partial f_4}{\partial w} + \frac{\partial f_5}{\partial v} + \frac{\partial f_6}{\partial u} \tag{3}$$

Note that the $\nabla \cdot f$ measure of the rat*e* at which volumes variation below the flow Φ_t of ***f***. Suppose D will be an area in R^6 with a sleek boundary and suppose D (t) = Ω_t (t), the image of D under $\Phi_t(t)$, the time t of the flow Φ_t of ***f***. Suppose V (t) indicates the volume of D(t). By Liouville's hypothesis, we obtain:

$$\frac{dV}{dt} = \int_{D(t)} (\nabla \cdot f)\, dxdydzdudw \tag{4}$$

And when you replace a value of $\nabla \cdot f$ in (4), the following equation is obtained:

$$\frac{dV}{dt} = (-a - h - 1 - k) \int_{D(t)} dxdydzdwdvdu \tag{5}$$

Where the Solving differential equation of first order (V),

V (t) is obtaining as: $v(t) = V(0)\, e^{(-a-h-1-k)t}$ (6)

"Equation 6" shows that any volume V (t) should be shrinking fast to zero exponentially with time →∞. Therefore, the new system described by "(1)" is a dissipative system, the all orbits of the "(1)" eventually confined to a specific of R^6, which has a zero volume. Thus, the asymptotic motion of the "(1)" is based on the attraction of the "(1)".

B. Symmetry

The new "(1)" is symmetry and invariant about the *z-axis* when coordinate (x , y , z , w , v , u) is transformed into (-x,-y , z , -w, -v, -u) , in order to prove that, let :

(x=-x, y=-y, z=z, w=w, v=v and u=u) (7)

And then we have:

$$-\frac{dx}{dt} = \frac{dx}{dt}, -\frac{dy}{dt} = \frac{dy}{dt}, \frac{dz}{dt} = \frac{dz}{dt}, \frac{dw}{dt} = \frac{dw}{dt}\quad \frac{dv}{dt} = \frac{dv}{dt},$$

and $\frac{du}{dt} = \frac{du}{dt}$. (8)

According to "(7)" and "(8)", the result is obtained as follows:

$$\begin{aligned}
-\frac{dx}{dt} &= ax - by - cw + dv \\
-\frac{dy}{dt} &= -ex + fxz + g\, e^{v} \\
\frac{dz}{dt} &= -hz + xy + iv \\
\frac{dw}{dt} &= -w - yz - gv \\
\frac{dv}{dt} &= jy + x - iz \\
\frac{du}{dt} &= -ku - jzw + lx
\end{aligned} \tag{9}$$

$$\begin{aligned}
\frac{dx}{dt} &= -ax + by + cw - dv \\
\frac{dy}{dt} &= ex - fxz - g\, e^{v} \\
\frac{dz}{dt} &= -hz + xy + iv \\
\frac{dw}{dt} &= -w - yz - gv \\
\frac{dv}{dt} &= jy + x - iz \\
\frac{du}{dt} &= -ku - jzw + lx
\end{aligned} \tag{10}$$

It is simple to notice that the "(1)" has invariant within the coordinates transformation (x, y, z, w, v, u) → (-x,-y, z, -w,-v,-u) which persists for all values of the system parameters. So, "(1)" has rotation symmetry about the z- axis, and u axis any non-trivial trajectory of the "(1)" It must have the twin trajectory. It means that this is a six-dimensional system that could also have symmetric pairs of coexisting attractors, such as limit cycles or strange attractors [14].

C. Equilibrium Point

We can get equilibrium points of the "(1)", nonlinear equations should be solved as follows:

$$\begin{aligned}
0 &= -ax + by + cw - dv \\
0 &= ex - fxz - g\, e^{v} \\
0 &= -hz + xy + iv \\
0 &= -w - yz - gv \\
0 &= jy + x - iz \\
0 &= -ku - jzw + lx
\end{aligned} \tag{11}$$

When a=9,b=11, = 0.1 , d=0.3 , e=30 , f=2.5 , g=5 , h=3, i= 0.5,j=4,k=15and l=25, it obtained three equilibrium points for the new hyper chaotic system as the following:

E0(x=0,y=0,z=0,u=0, w=0) , E1(x=0.0539309, y=-0.0367507 , z=-0.186144 , w=5.55766 , v=-1.1129, u=0.365758) ,when the linear "(1)" around the equilibrium point E*, with a view to determining the Jacobian matrix we get:

$$f = \begin{cases}
f_1 = \frac{dx}{dt} = -ax + by + cw - dv \\
f_2 = \frac{dy}{dt} = ex - fxz - g\, e^{v} \\
f_3 = \frac{dz}{dt} = -hz + xy + iv \\
f_4 = \frac{dw}{dt} = -w - yz - gv \\
f_5 = \frac{dv}{dt} = jy + x - iz \\
f_6 = \frac{du}{dt} = -ku - jzw + lx
\end{cases} \tag{12}$$

$$J=\begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} & \frac{\partial f_1}{\partial w} & \frac{\partial f_1}{\partial v} & \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} & \frac{\partial f_2}{\partial w} & \frac{\partial f_2}{\partial v} & \frac{\partial f_2}{\partial u} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} & \frac{\partial f_3}{\partial w} & \frac{\partial f_3}{\partial v} & \frac{\partial f_3}{\partial u} \\ \frac{\partial f_4}{\partial x} & \frac{\partial f_4}{\partial y} & \frac{\partial f_4}{\partial z} & \frac{\partial f_4}{\partial w} & \frac{\partial f_4}{\partial v} & \frac{\partial f_4}{\partial u} \\ \frac{\partial f_5}{\partial x} & \frac{\partial f_5}{\partial y} & \frac{\partial f_5}{\partial z} & \frac{\partial f_5}{\partial w} & \frac{\partial f_5}{\partial v} & \frac{\partial f_5}{\partial u} \\ \frac{\partial f_6}{\partial x} & \frac{\partial f_6}{\partial y} & \frac{\partial f_6}{\partial z} & \frac{\partial f_6}{\partial w} & \frac{\partial f_6}{\partial v} & \frac{\partial f_6}{\partial u} \end{bmatrix} \quad (13)$$

$$J=\begin{bmatrix} -a & b & 0 & c & -d & 0 \\ e-fz & 0 & -fx & 0 & -ge^{v} & 0 \\ y & x & -h & 0 & i & 0 \\ 0 & -z & -y & -1 & -g & 0 \\ 1 & j & -i & 0 & 0 & 0 \\ l & 0 & -jw & -jz & 0 & -k \end{bmatrix} \quad (14)$$

So for equilibrium poinE0(x=0,y=0,z=0,u=0,w=0), and a=9 , b=11 ,c=0.1_d=0.3, e=30, f=2.5,g=5,h=3 , i=0.5, j=4, k=15 and l=25, the result of Jacobian matrix has the following result:

$$J=\begin{bmatrix} -9 & 11 & 0 & 0.1 & -0.3 & 0 \\ 30 & 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & -3 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & -1 & -5 & 0 \\ 1 & 4 & -0.5 & 0 & 0 & 0 \\ 25 & 0 & 0 & 0 & 0 & 15 \end{bmatrix} \quad (15)$$

To obtaining the eigenvalues, assume |λI-J0|=0, then the eigenvalues that identical to equilibrium E0 (*0, 0, 0, 0, 0, 0*) are respectively get as following: $\lambda_1=$-22.9838, $\lambda_2=$-15, $\lambda_3=$ 13.0759, $\lambda_4=$-2.92751, $\lambda_5=$ -1.10293 and $\lambda_6=$0.938297.

Therefore, the equilibrium E0 (0,0,0,0,0,0) is considered as a saddle point, and the hyper chaotic system it could be considered unstable in this point *E*1. At the one time, it is easy to prove that the equilibrium point *E*1 is too unstable saddle points. To the equilibrium point E1(*x=0.0539309, y=0.0367507,z=0.186144,w=5.55766,v=1.1129,u=0.365758*),and a=9 ,b=11, c=0.1 ,d=0.3, e=30, f=2.5 ,g=5 , h=3 ,i=0.5 ,j=4, k=15,and l=25 , The result of the Jacobian matrix would be:

$$J=\begin{bmatrix} -9 & 11 & 0 & 0.1 & -0.3 & 0 \\ 30-2.5*-0.186 & 0 & -2.5*0.053 & 0 & -5e^{-1.112} & 0 \\ -0.036 & 0.053 & -3 & 0 & 0.5 & 0 \\ 0 & 0.186 & -1 & -5 & 0 & 0 \\ 1 & 4 & -0.5 & 0 & 0 & 0 \\ 25 & 0 & -4*5.557 & 4*0.186 & 0 & -15 \end{bmatrix} \quad (16)$$

In the same method, the eigenvalues that corresponding to equilibrium point E1 they were obtained as: -23.295, $\lambda_2=$-15, $\lambda_3=$ 13.9395, $\lambda_4=$-2.92163, $\lambda_5=$-1.13428, $\lambda_6=$ 0.411404.Therefore, the equilibrium E1 (0.0539309, -0.0367507,-0.186144, 5.55766,-1.1129, 0.365758) is considered as a saddle point. So, the hyper-chaotic system is unstable onto the point *E*1 [15].

D.Lyapunov Exponents and Lyapunov Dimensions.

The new Six-Dimension hyper chaotic system has six Lyapunov exponents they are obtained as follows: L1= 1.04609, L2= 0.121133, L3= 0.0260565, L4= -1.10258, L5= -13.0445 and L6= -15.0464 when the values of control parameters and the initial condition of the "(1)" , selected as When a=9,b=11,c=0.1,d=0.3,e=30,f=2.5,g=5,h=3, i=0.5,j=4,k=15 and l=25 and x(0)=0.1, y(0)=0.5, z(0)=3.5,w(0)=0.6. v (0)=0.4 and u(0)=0.1 , we can be seen that the Maximal Lyapunov exponent is positive which it L1= 1.04609 , Indicates that the new system has properties chaos. Since it has three positive values of Lyapunov exponents and the rest three values of Lyapunov exponents are negative, Since L1+L2+ L3 > 0 and L1+L2+L3 < 0. Consequently, this system is hyper chaotic. The fractional dimension is as well a typical feature of chaos calculated Kaplan-York dimension via the Lyapunov exponents, and *D*KY for the new system could be obtained as follows [16]:

$$D_{KY}=j+\frac{1}{|L_{j+1}|}\sum_{i=1}^{j}L_i$$

$$D_{KY}=4+\frac{1}{|L_{j+1}|}\sum_{i=1}^{4}L_i=4+\frac{L_1+L_2+L_3+L_4}{L_5}$$

$$=4+\frac{1.04608+0.12113+0.02605+-1.10257}{13.0445}=4.00695$$

This Lyapunov dimension of a new hyper chaotic system of the "(1)" is fractional nature indicated that the new system has non-periodic orbits and that refers to nearby trajectories diverge. Thus, there is truly chaos in this chaotic system.

E. Waveform Analysis of the New Chaotic System

As is known the waveforms of an "(1)" (x(t),y(t),z(t),w(t),v(t),u(t)) in the time domain are shown in Figs. 9 and 10 . Evidently, the time domain waveform for the new system has a non- cyclical feature, in respect of recognizing between a different periodic motion that can too appear a complicated behavior and a chaotic motion, that it can be observed that the waveform of the time domain has non-periodic properties. It is one of the chaotic system properties.

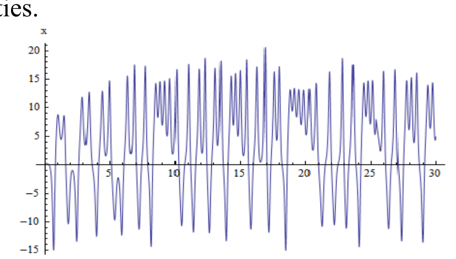

Fig.9. Time versus (*x*) of the novel chaotic system

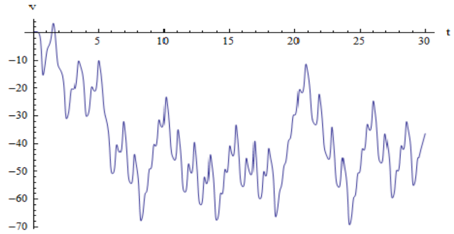

Fig.10. Time versus (v) of the novel chaotic system.

F. Sensitivity for Initial Conditions

Unpredictability is the most characteristic property of a chaotic system and that because of the sensitive dependence of solutions on initial conditions. Two different initial conditions, regardless of how close they are, will be broadly separated and therefore, for any given number of precision numbers in an initial condition, there will be the time in the future when accurate forecasts of the state of the system cannot be made. Figs. 11 and 12 presented the development of chaos trajectories is highly sensitive to the initial conditions. System initial values are set to: x(0) =0.1,y(0) =0.5, z(0) = 3.5, w(0)=0.6, v(0)=0.4 and u(0)=0.1 to the solid line and x(0)= 0.00000000001, y(0)=0.5,z(0)= 3.5, w(0)=0.6. v(0)=0.4 and u(0) = 0.1, to the dashed line.

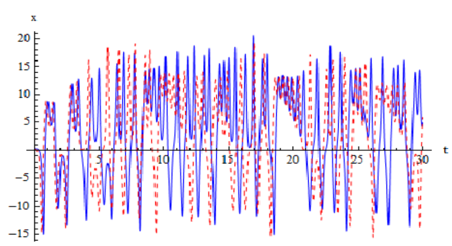

Fig.11. Sensitivity tests of a novel system x(t)

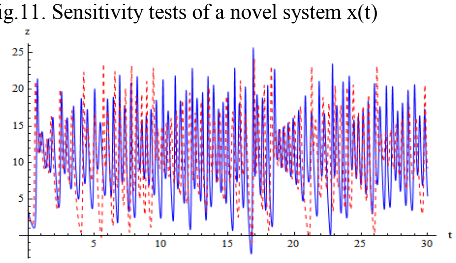

Fig.12. Sensitivity tests of a novel system z(t)

IV. CONCLUSION

The new Six-Dimension hyper chaotic system offered above Contains complicated dynamics behaviors and all properties of the chaotic system, and it has the six Lyapunov exponents. The proposed system with different topology and structure than the existing six-dimension systems is introduced in this paper .The basic characteristics and dynamic behavior of the new system have been examined to prove it's chaotic. The new system generates hyper chaotic behavior when selecting the parameters as: a∈[9,9.5],b∈[11,11.5],c∈[0.1,0.5] ,d∈[0.3,0.8] ,e∈[30,30.5], f=2.5 , g∈[5,5.5], h∈[3,3.5], i∈ [0.5,1], j∈[4,4.5],k∈[15,15.5] and l∈[25,25.5] and the initial conditions set as the following: x(0)=0.1, y(0)=0.5, z(0) =3.5 , w (0) =0.6. v(0) = 0.4 and u (0) =0.1. The new 6-D hyper-chaotic system contains three unstable equilibrium points and the Lyapunov Exponents of the new system are calculated as follows :L1= 1.04609, L2= 0.121133, L3= 0.0260565, L4= -1.10258, L5= -13.0445 and L6= -15.0464 this means a system is hyper-chaotic because the new system contains three Lyapunov exponents positive, the fractal dimension is 4.00695 and new system characteristics with high sensitivity and generates a complex chaotic attractor. In the future work, we recommend using this six-dimensional system to generate random keys for the purpose of data encryption.

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